Report on the Practical Assignment in DGSP

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March 12, 2026

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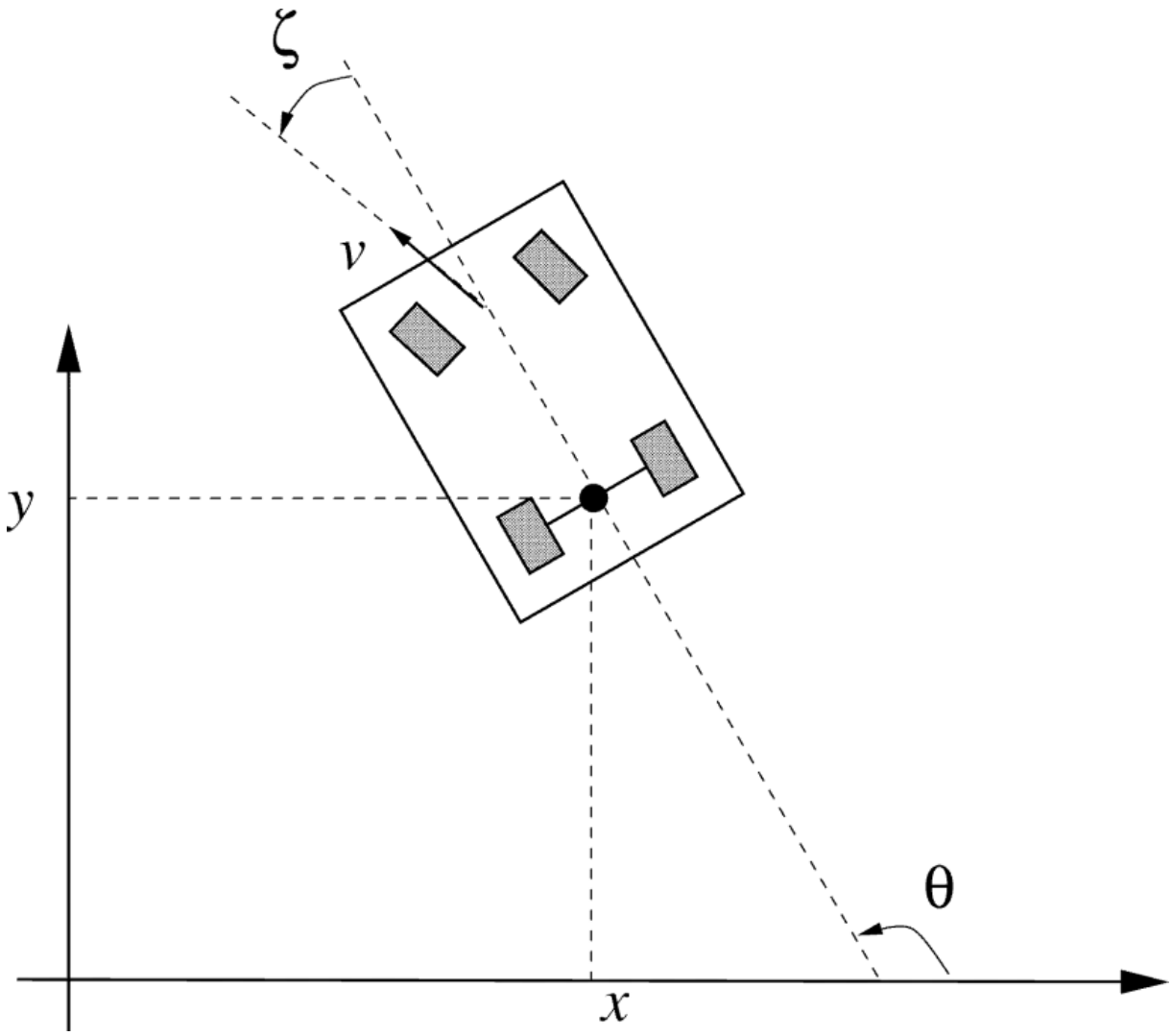
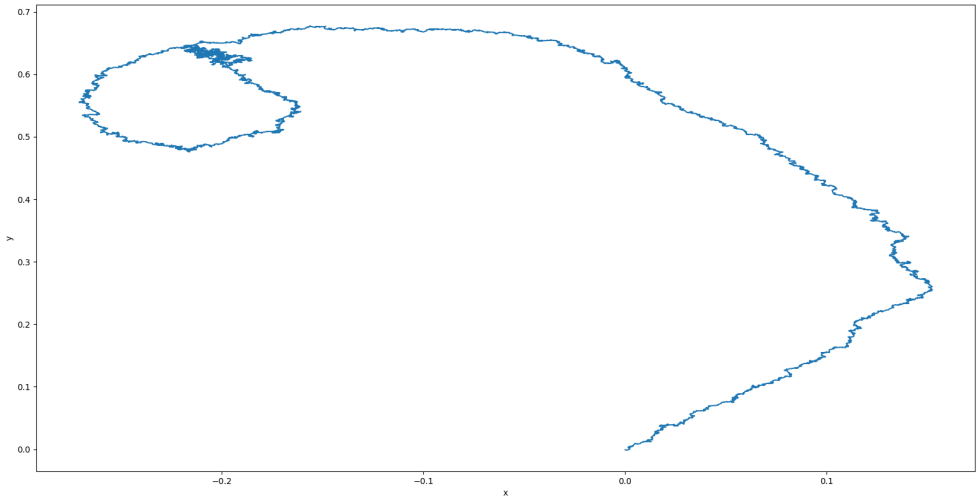
1 Theoretical Part

1.1 Phenomenon Under Study

This paper considers the problem of observing a cart moving on a plane. The state of the cart is given by its position on the plane (x, y, orientation in space) and the steering angle of the wheels. A locator is placed at the origin of the coordinate system; it measures the distance to the cart and the direction angle toward it.



Next, the mathematical model used in this work will be described. A typical trajectory obtained with its help looks as follows:



1.2 Observation System

The mathematical model is described by a discrete-time system:

$$\begin{cases} x_{1,k} = x_{1,k-1} + Tcos(\theta_{k-1})cos(\phi_{k-1})u_{1,k-1} \\ x_{2,k} = x_{2,k-1} + Tsin(\theta_{k-1})cos(\phi_{k-1})u_{1,k-1} \\ \theta_k = \theta_{k-1} + Tsin(\phi_{k-1})\frac{u_{1,k-1}}{l} + \omega_{3,k} \\ \phi_k = \phi_{k-1} + Tu_{2,k-1} + \omega_{4,k} \end{cases}$$

where

- x_1 is the object's coordinate along the x-axis
- x_2 is the object's coordinate along the y-axis
- θ is the angle between the cart's direction of motion and the positive semi-axis Ox (orientation on the plane)
- ϕ is the steering angle of the cart's wheels relative to the direction of motion
- T is the time discretization parameter of the system
- l is the distance between the cart's axles
- ω is the noise in the dynamics model, $\omega_k \sim \mathcal{N}(0, Q)$
- u_1 is the controlled linear velocity
- u_2 is the controlled angular velocity

The observations are defined as follows:

$$\begin{cases} r_k = \sqrt{x_{1,k}^2 + x_{2,k}^2} + \nu_{1,k} \\ \alpha_k = \arctan(\frac{x_{2,k}}{x_{1,k}}) + \nu_{2,k} \end{cases}$$

where

- r_k is the distance from the locator to the target
- α_k is the direction angle from the locator to the target
- ν is the observation noise, $\nu \sim \mathcal{N}(0, R)$

2 Practical Part

2.1 Values of the System Parameters

In the numerical experiments, we set the following system parameters:

$$l = 0.1,\ u_1 = 3,\ u_2 = 0$$

Time steps:

$$\delta_1 = \delta_2 = 10^{-3},\ \delta_1 = 10^{-2},\ T_{max} = 1$$

Note that the equality δ_1 and δ_2 follows from the fact that the system has already been discretized in advance.

Also, for the observations we will use $\arctan_2$ instead of arctan. This is due to the fact that the domain of arctan is $[-\frac{\pi}{2}; \frac{\pi}{2}]$, whereas $\arctan_2$ covers the entire trigonometric circle. This allows us to avoid observation errors caused by an incorrect determination of the half-plane.

We take the noise covariance matrices as follows:

$$Q = \begin{bmatrix} 3 \cdot 10^{-7} & 0 & 0 & 0 \\ 0 & 3 \cdot 10^{-7} & 0 & 0 \\ 0 & 0 & 3 \cdot 10^{-5} & 0 \\ 0 & 0 & 0 & 3 \cdot 10^{-5} \end{bmatrix}$$

$$R = 5 \cdot 10^{-3} \cdot I_2$$

The initial value of the trajectory is modeled as follows:

$$X_0 \sim \mathcal{N}(\mu, 0.3 \cdot I_4)$$

where $\mu = (0, 0, \frac{\pi}{4}, 0)$

All parameters required for filtering were computed analytically during the program's execution using the corresponding Python packages (filterpy, sympy). Implementations of all filters were also taken from the corresponding Python packages. In the particle filter, the number of particles is 1000. To achieve higher precision, 128-bit floating-point numbers were used (IEEE 754 standard).

All computations were carried out on a bundle of 10,000 trajectories.

2.2 Comparison of Algorithm Performance

2.2.1 Trajectory Estimates

Figure 1: Estimate of x_1

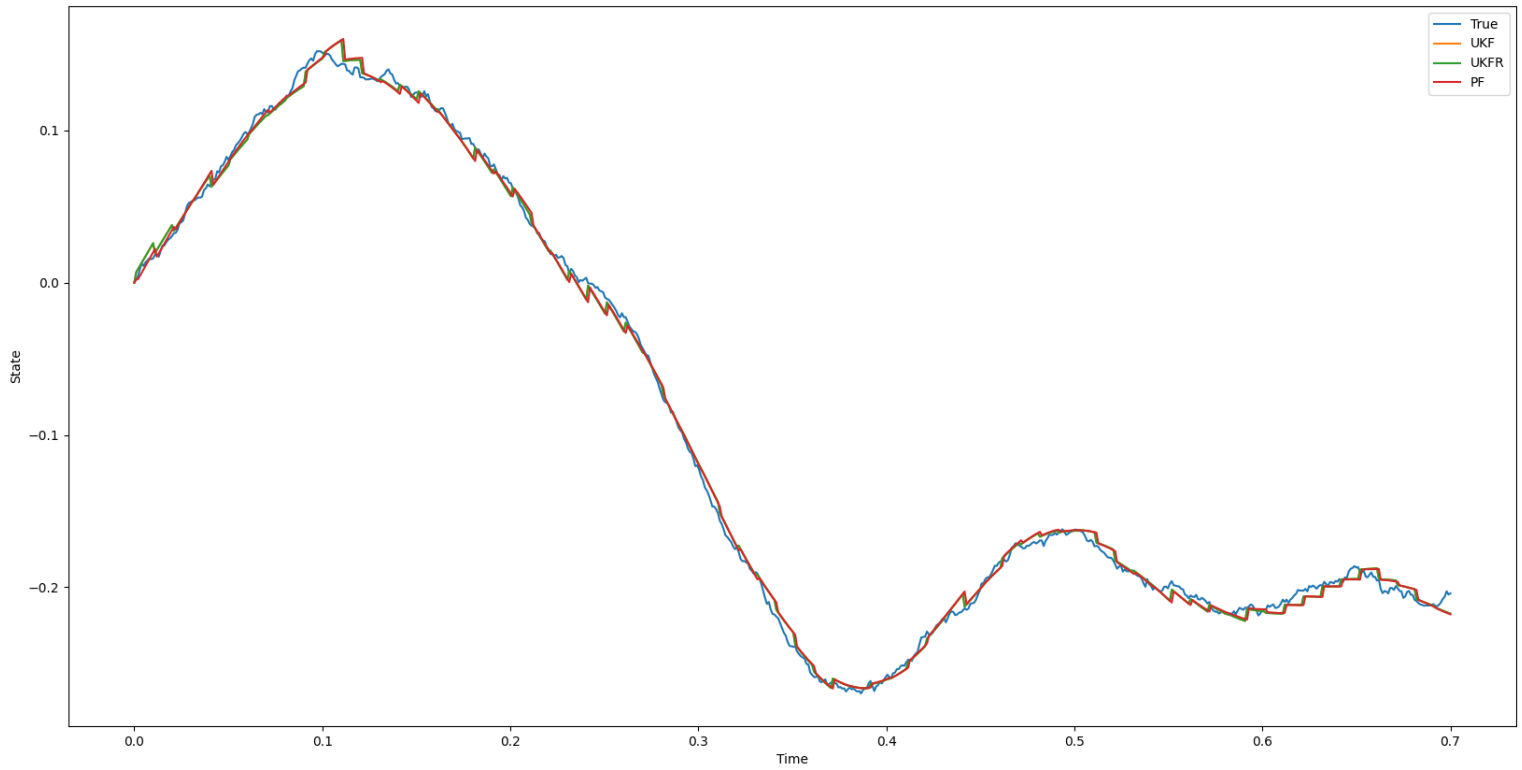


Figure 2: Estimate of x_2

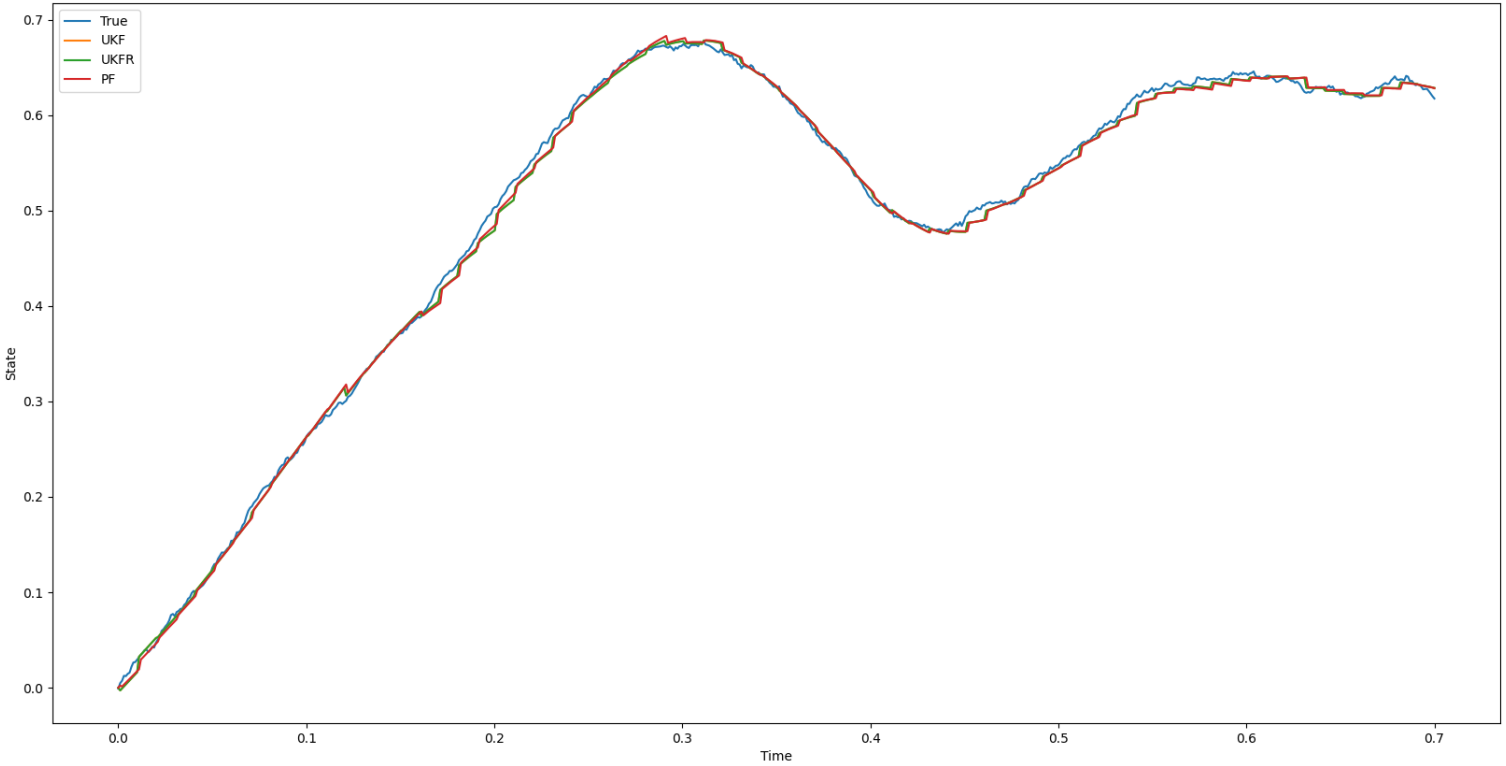


Figure 3: Estimate of θ

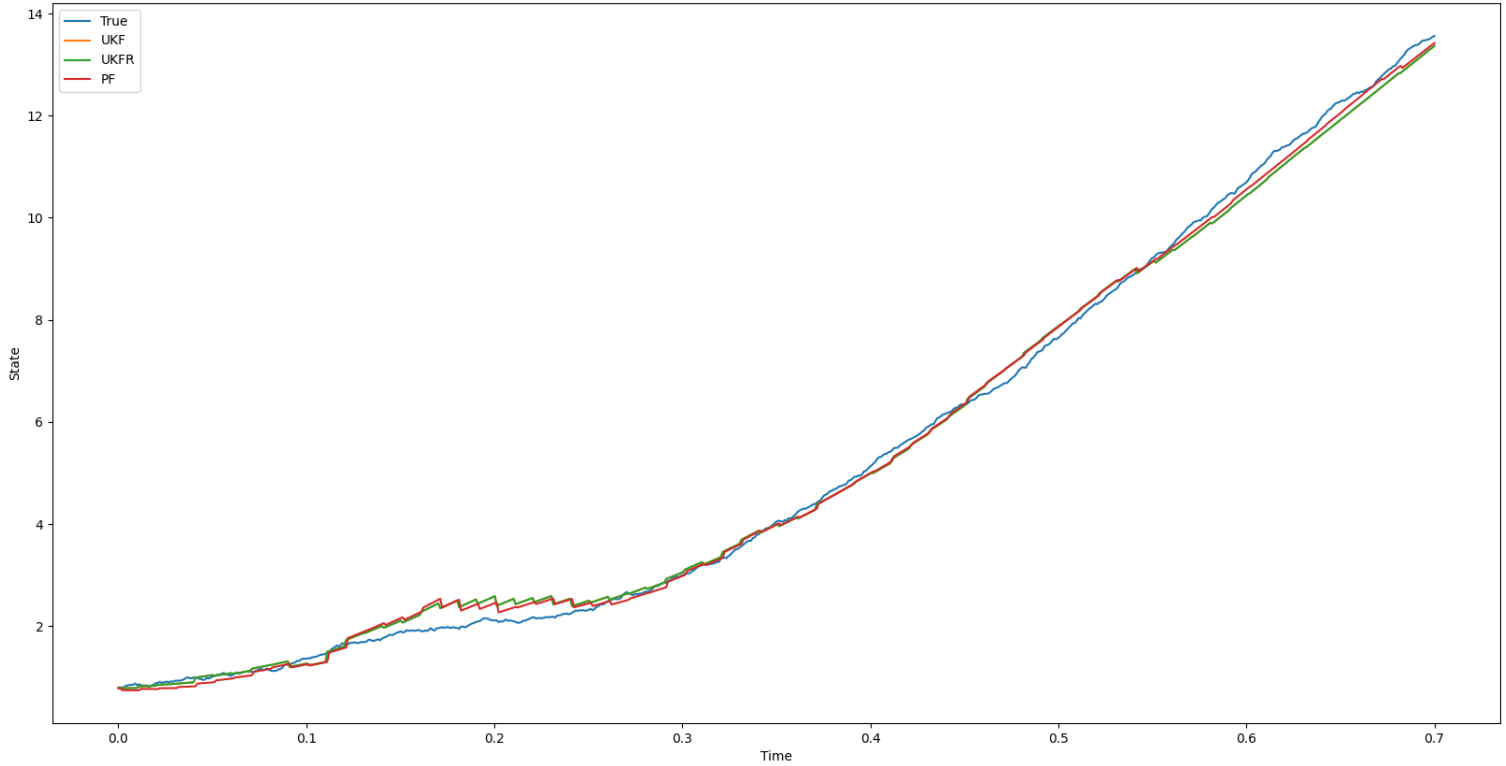
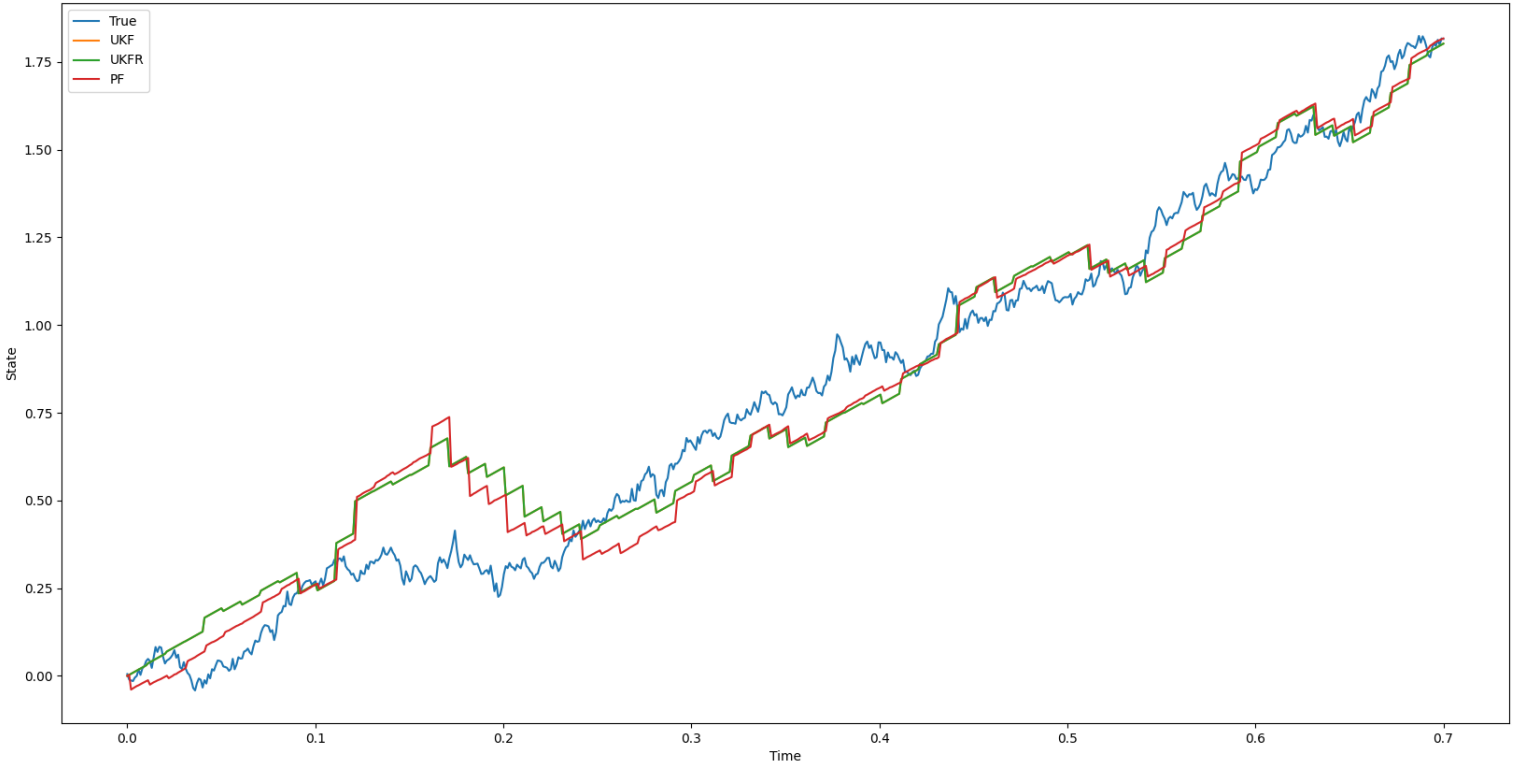
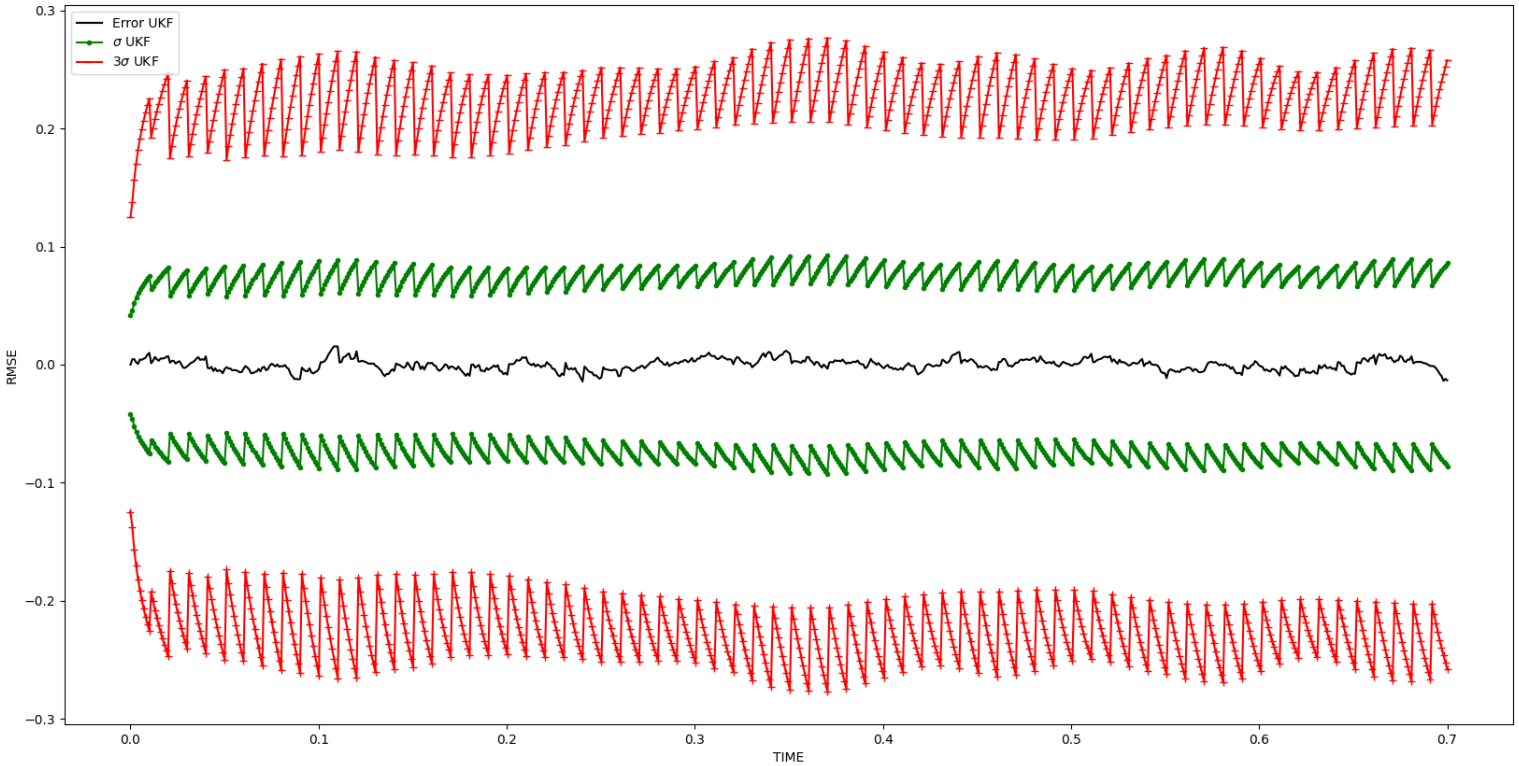


Figure 4: Estimate of ϕ

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RMSE
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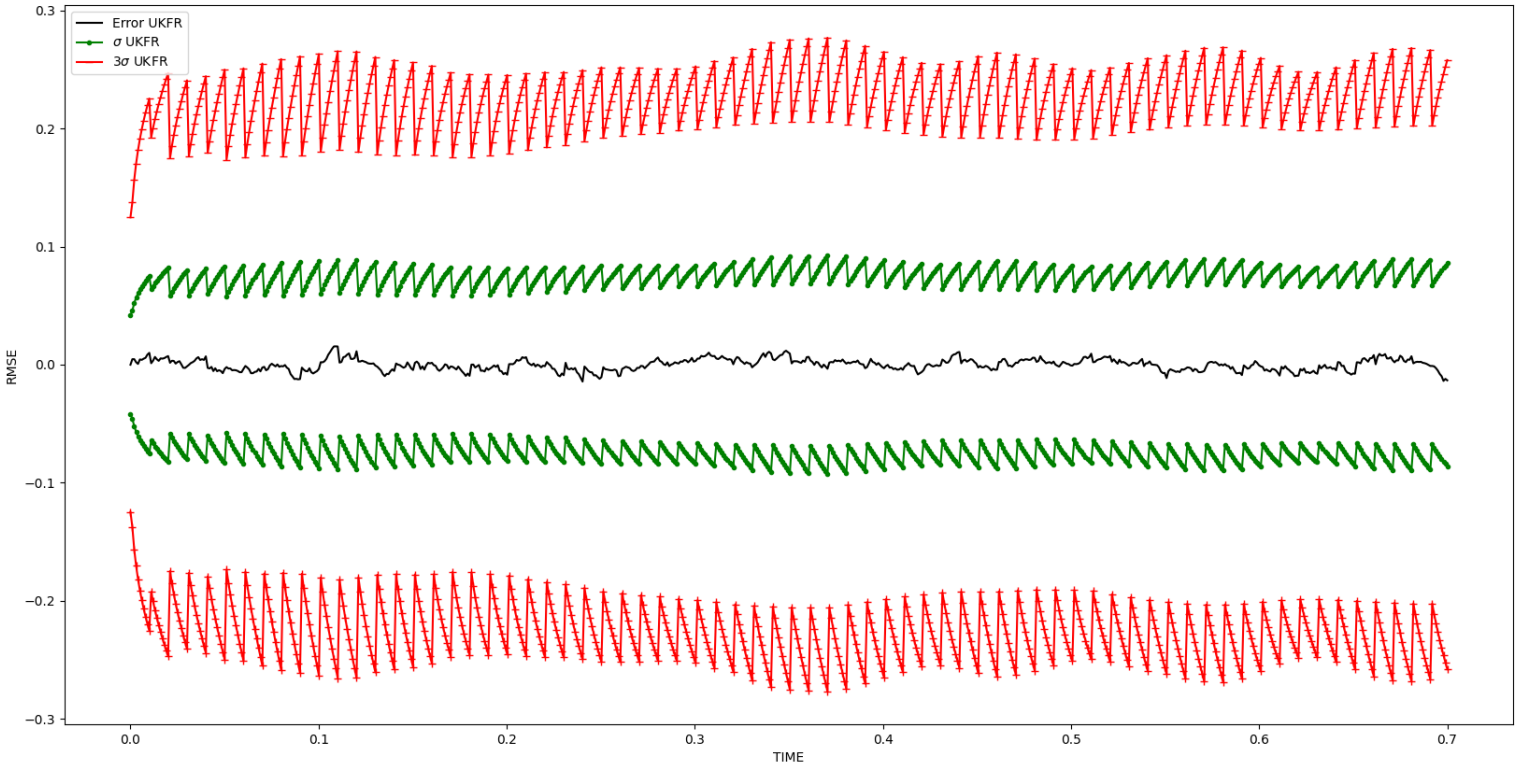
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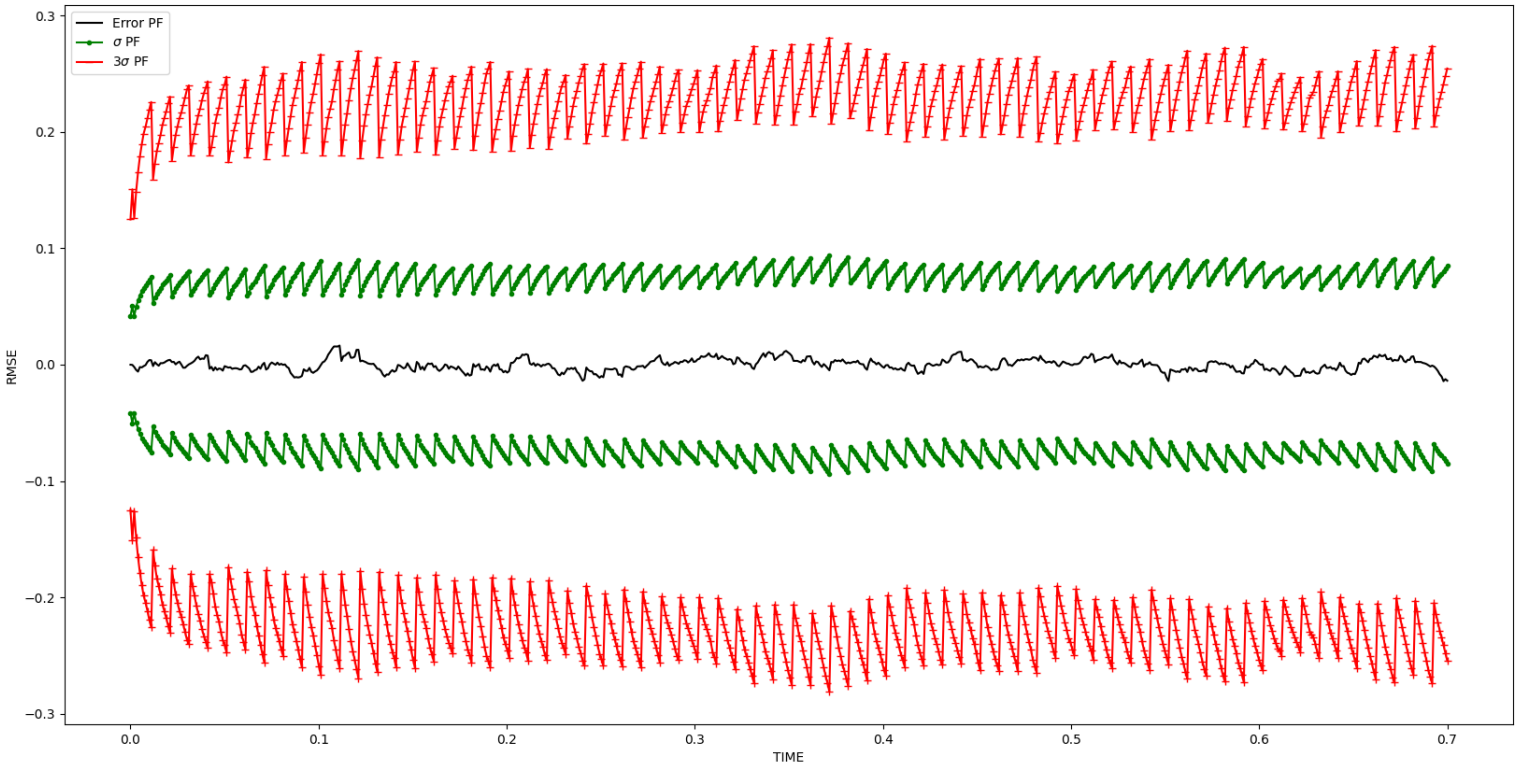
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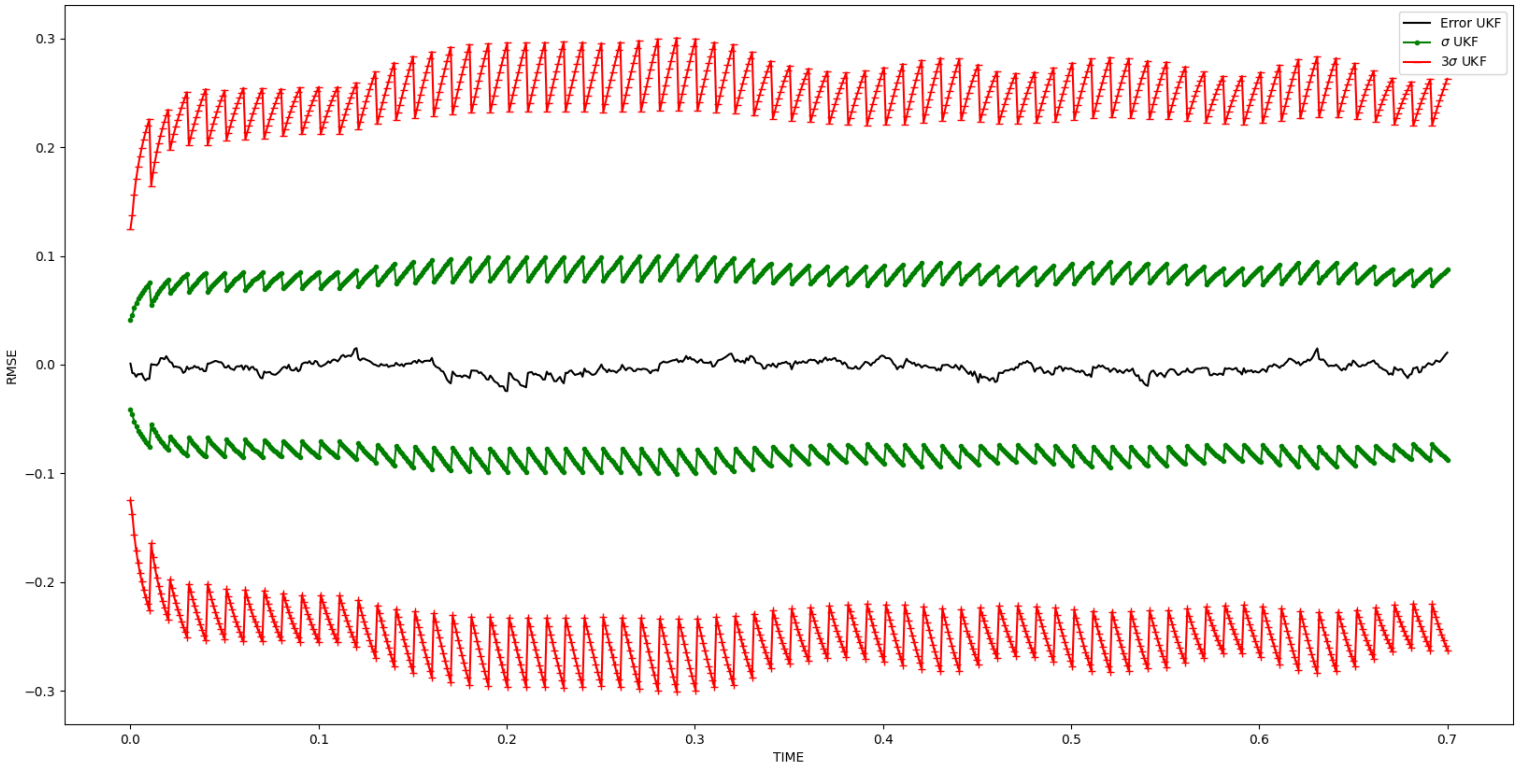
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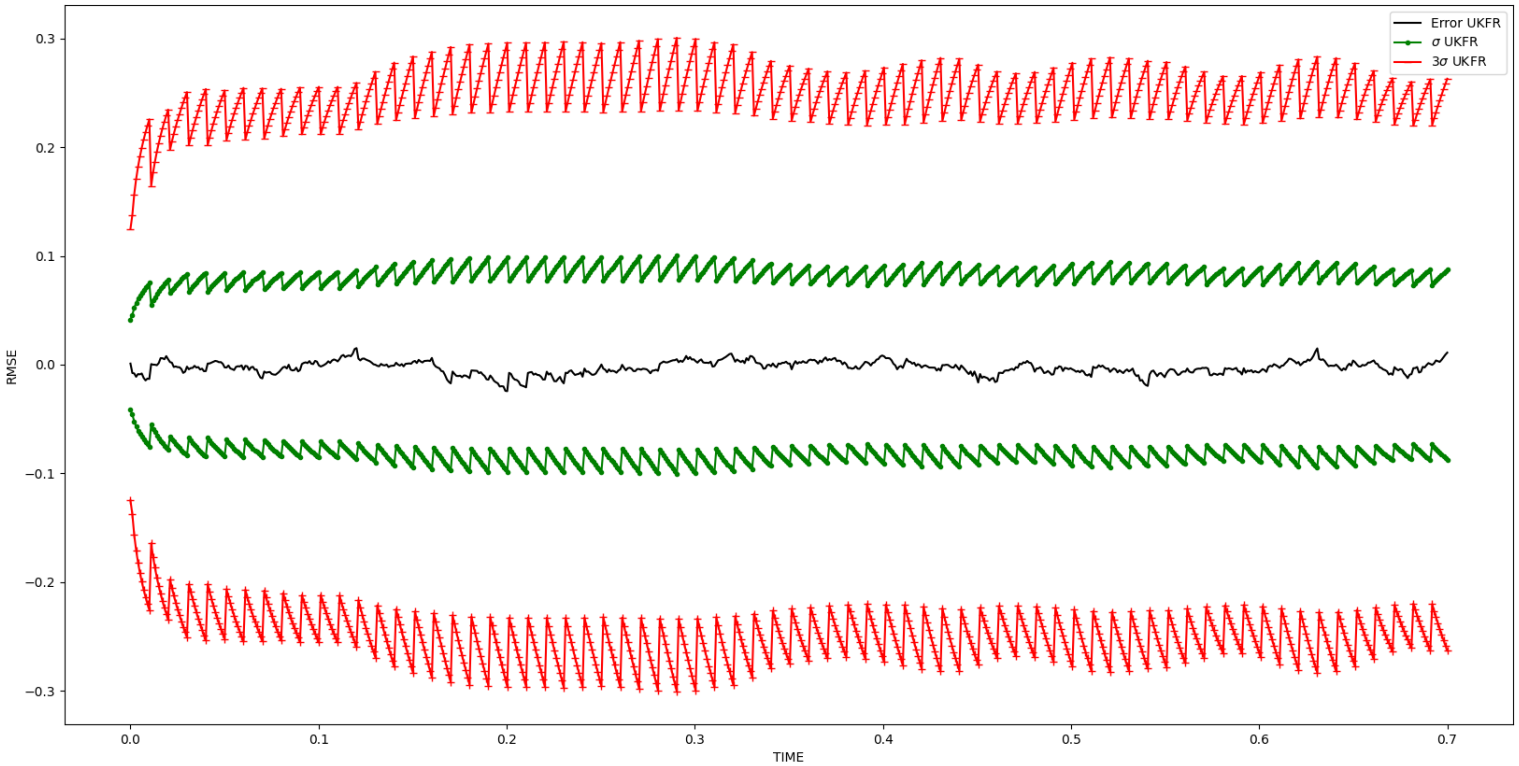
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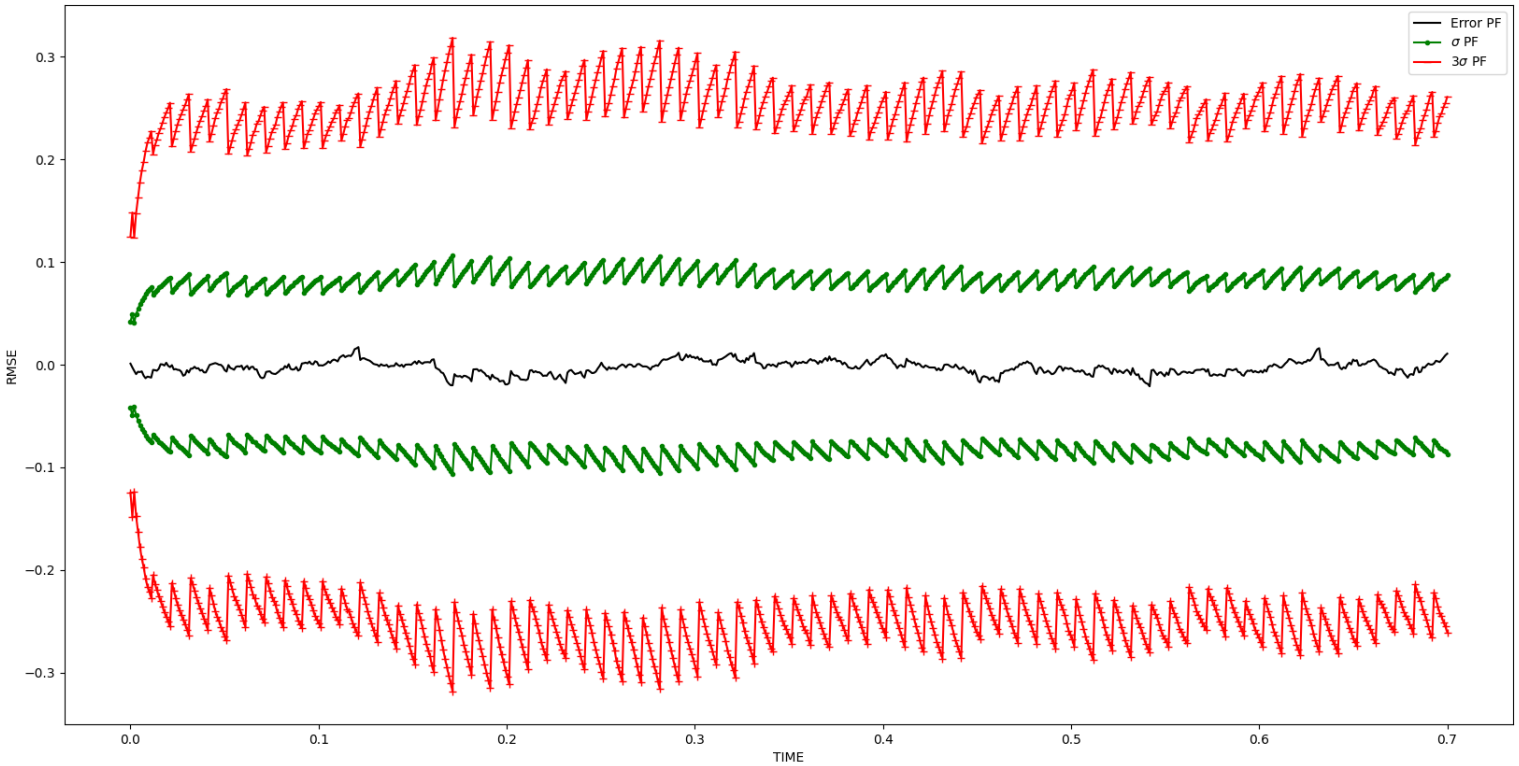
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Error UKF
σ UKF
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Error UKFR
σ UKFR
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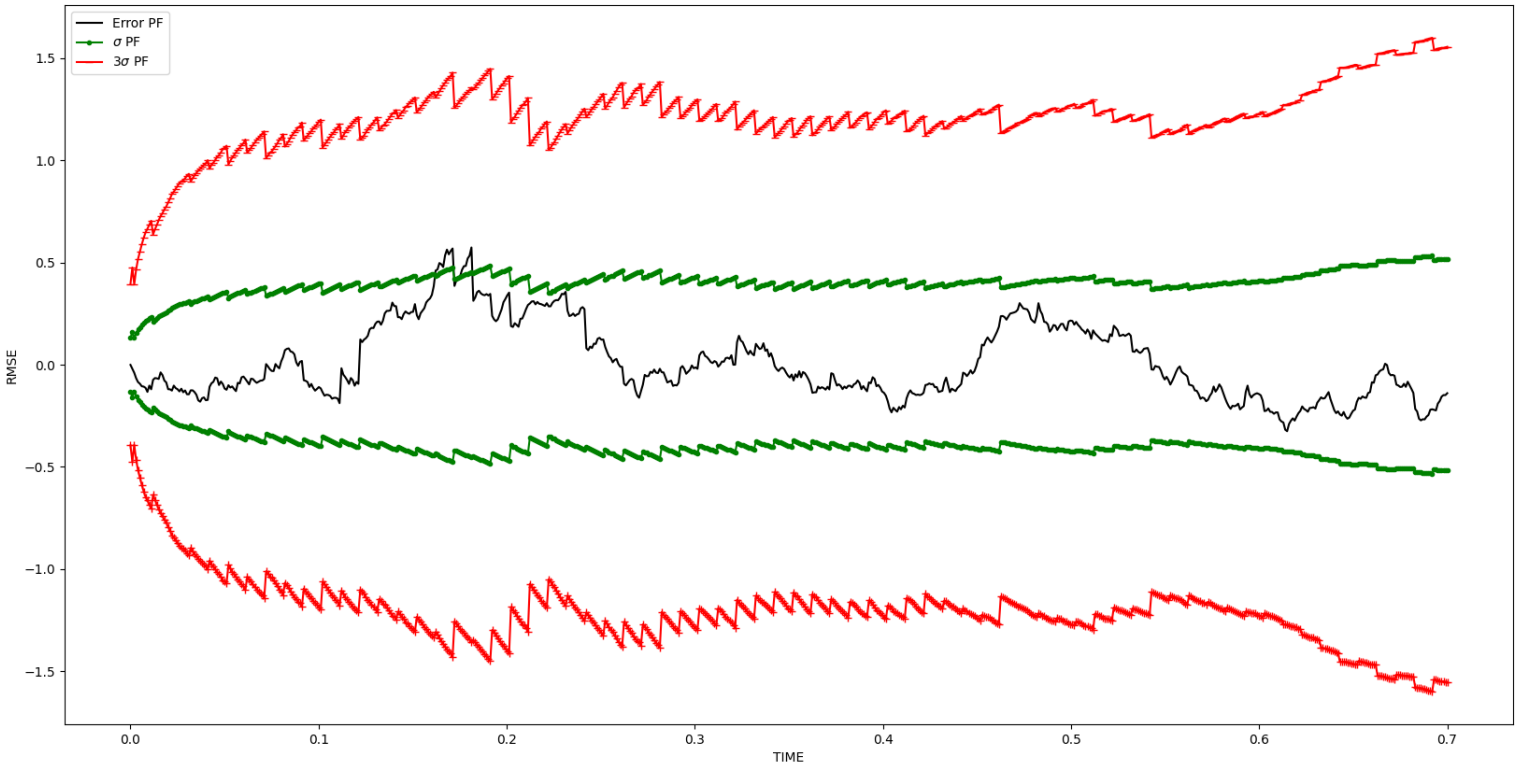
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0.7

Error UKF
σ UKF
3σ UKF
1.0
0.5
0.0
−0.5
−1.0
RMSE
0.0
0.1
0.2
0.3
0.4
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TIME

Error UKFR
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3σ UKFR
1.0
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0.0
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RMSE
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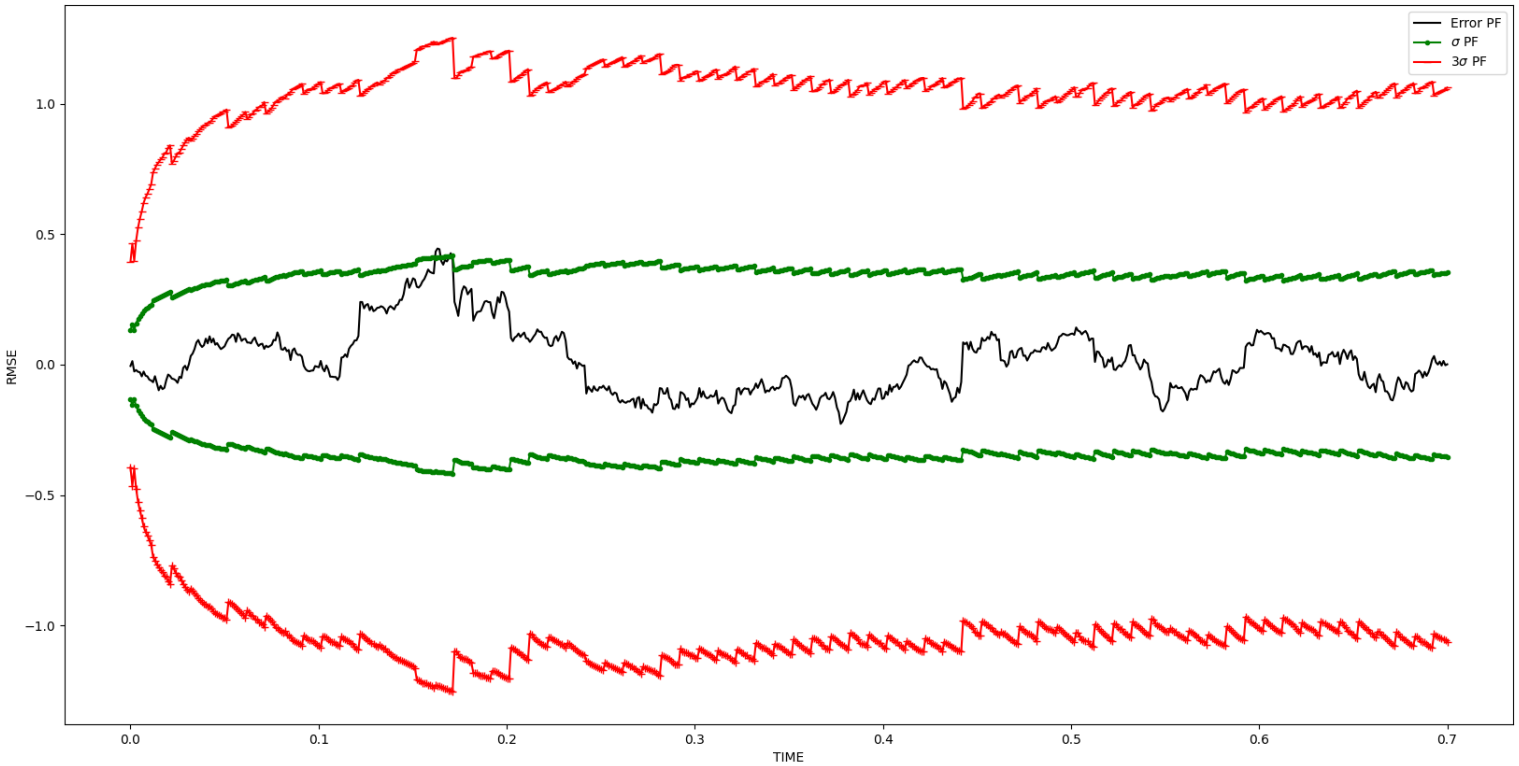
TIME


Error PF
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0.5
0.6
0.7

2.2.2 Error Plots for Non-Divergent Trajectories

Figure 5: Error of x_1

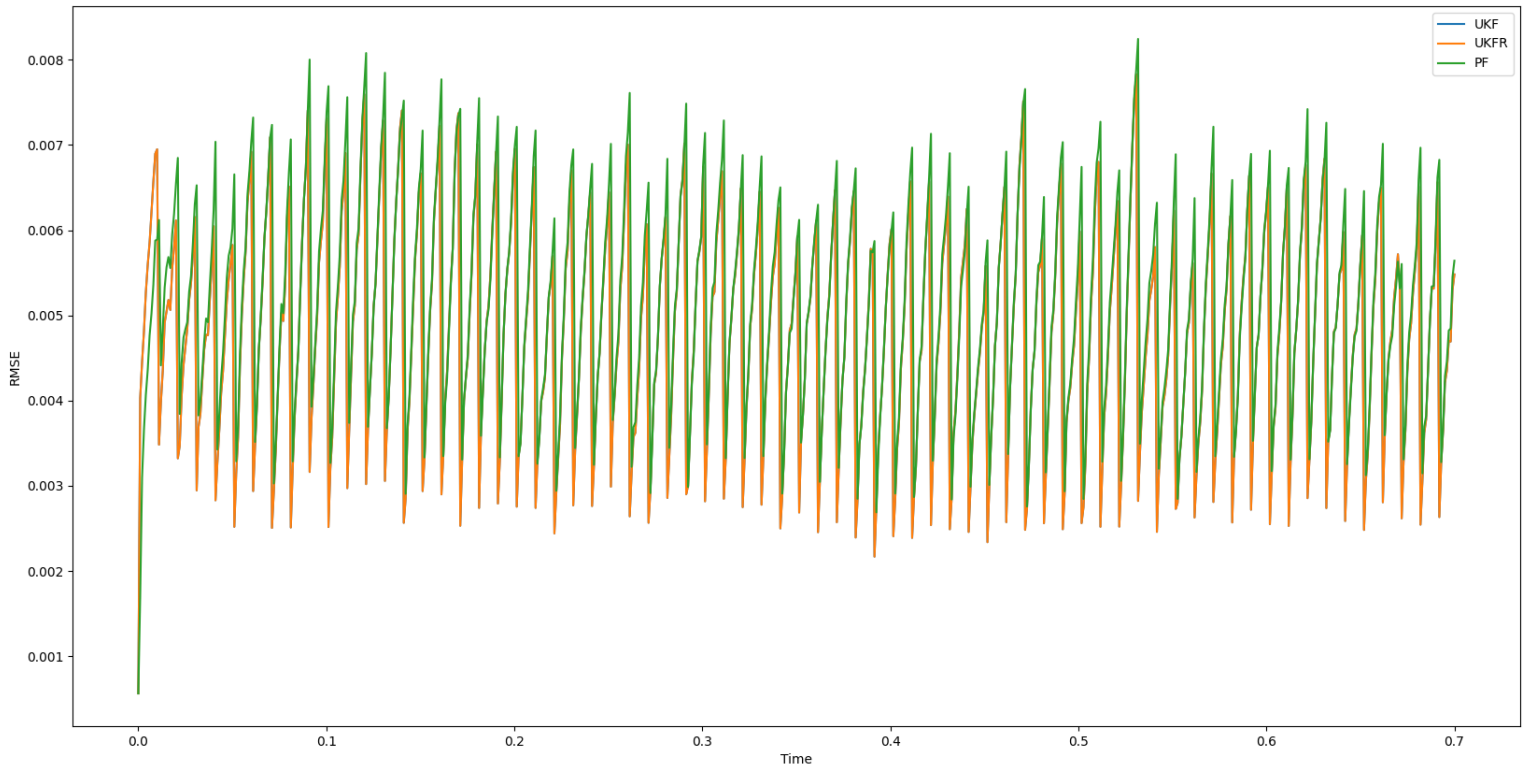


Figure 6: Error of x_2

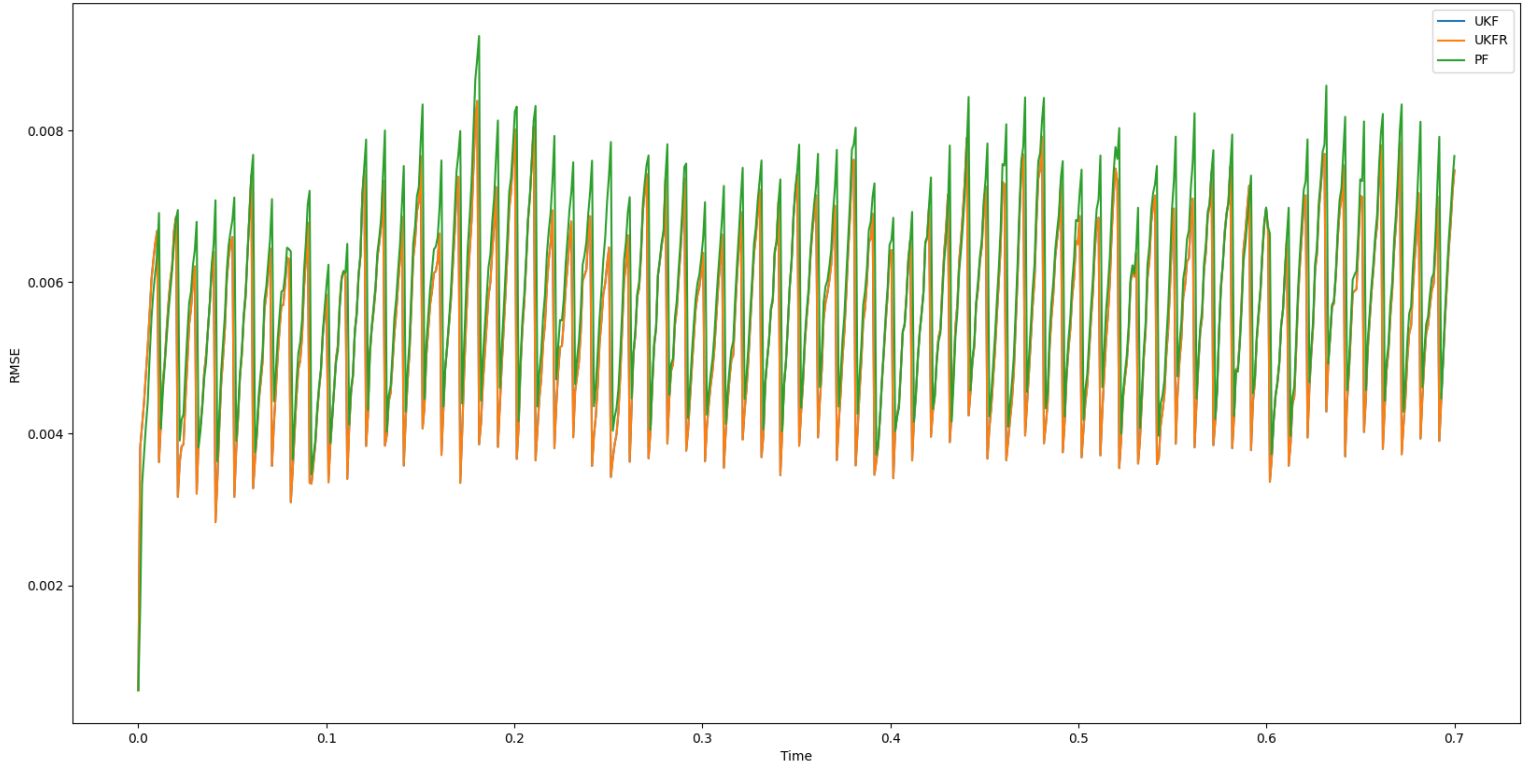


Figure 7: Error of θ

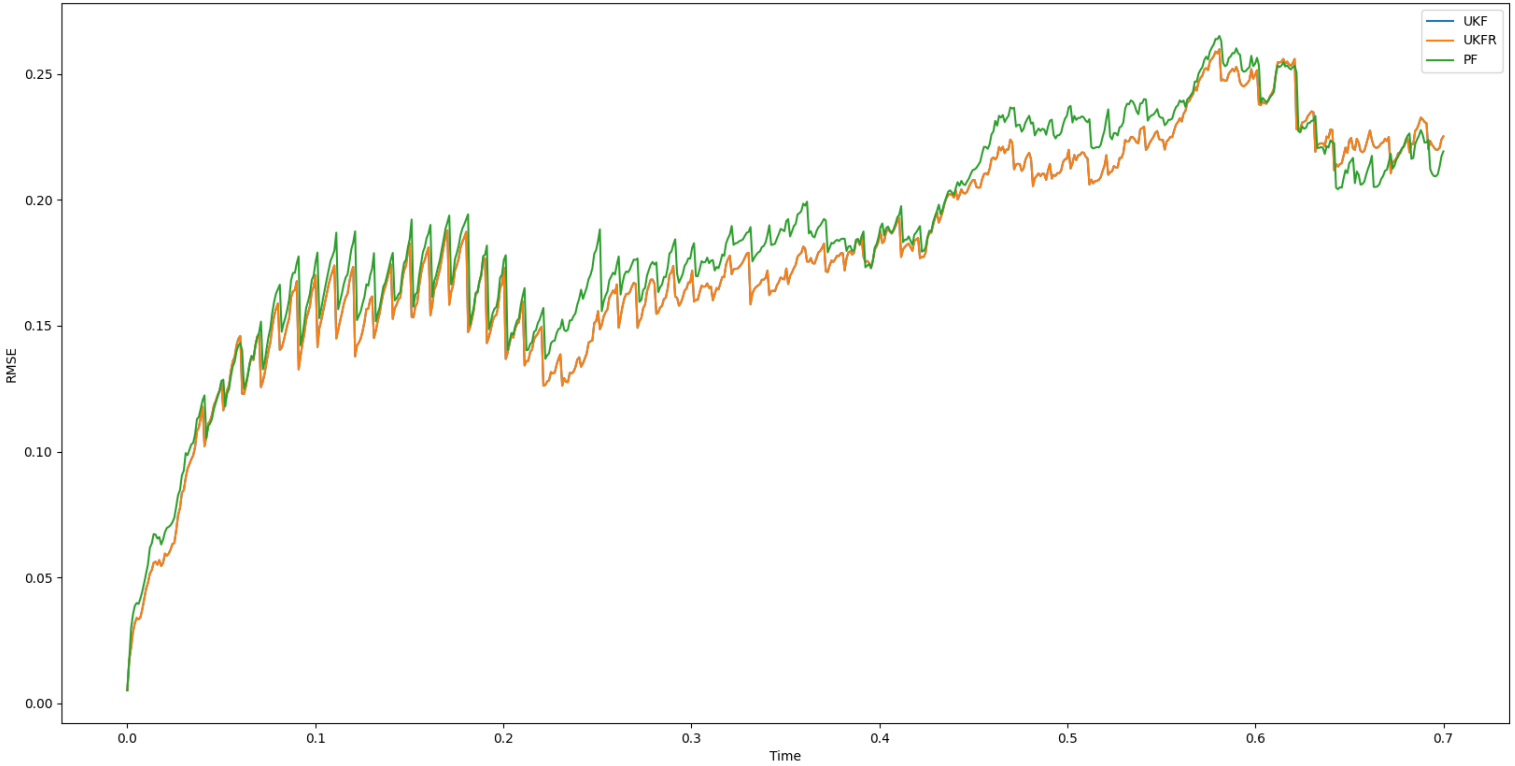
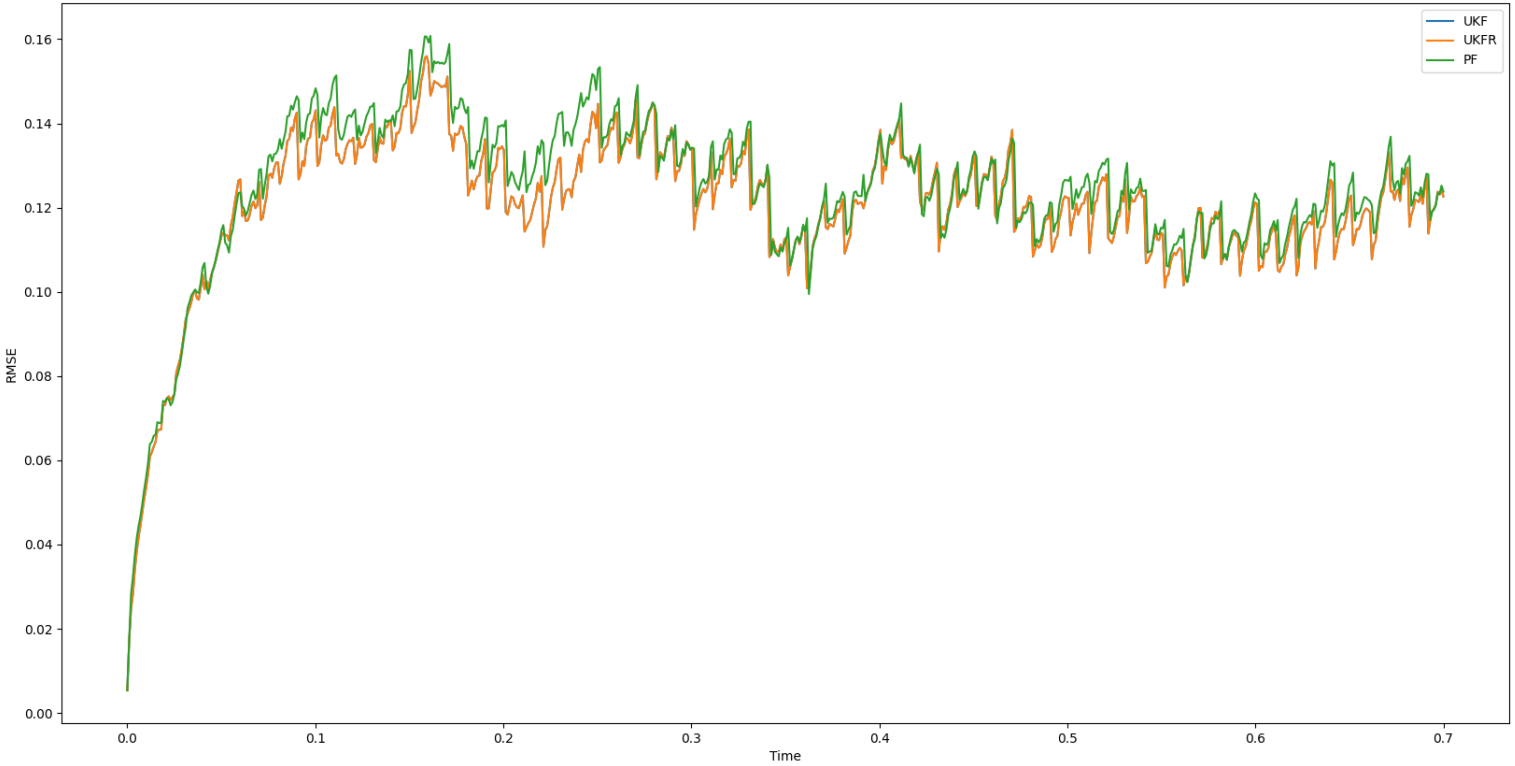


Figure 8: Error of ϕ



2.2.3 Error Plots for All Trajectories

Figure 9: Error of x_1

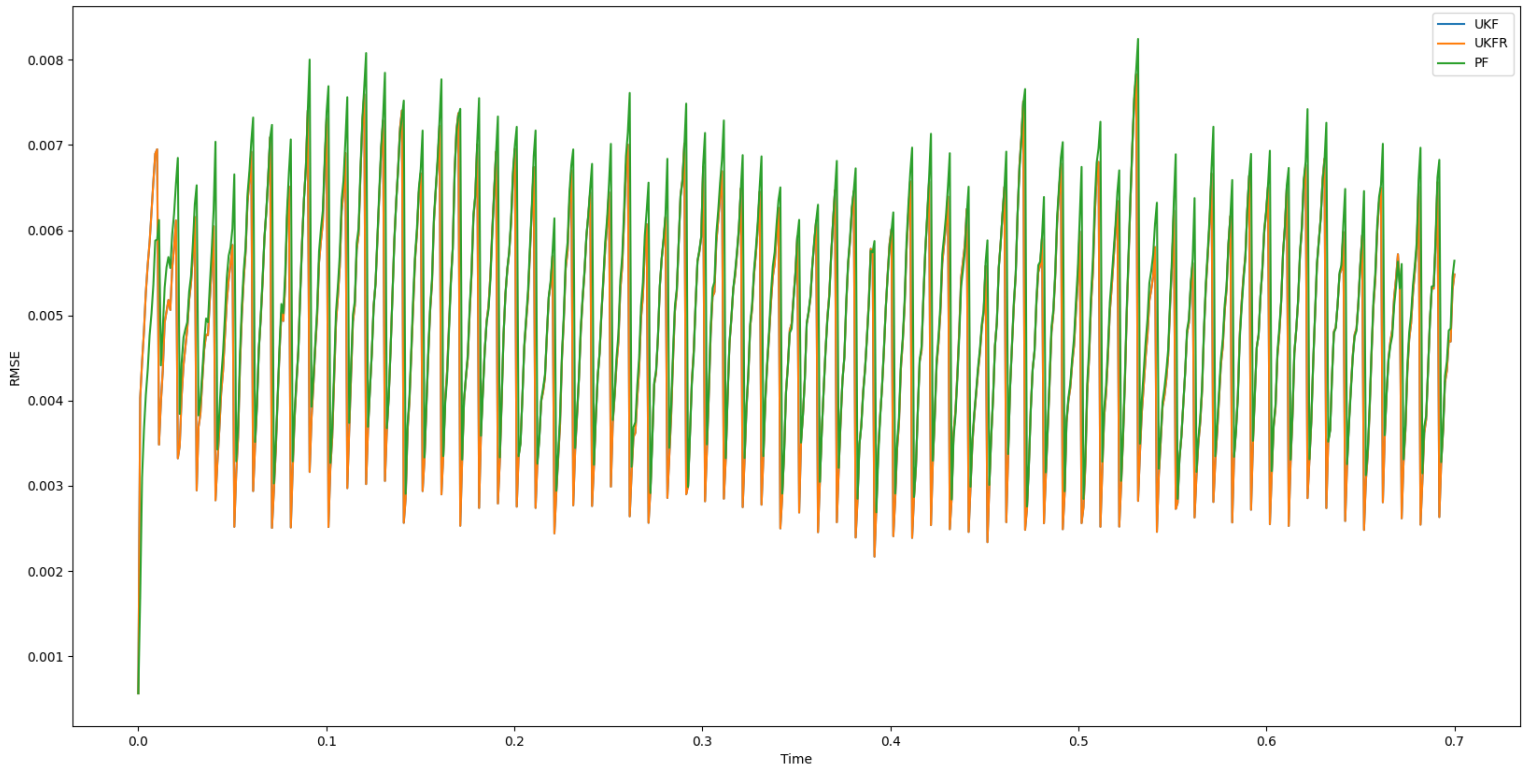


Figure 10: Error of x_2

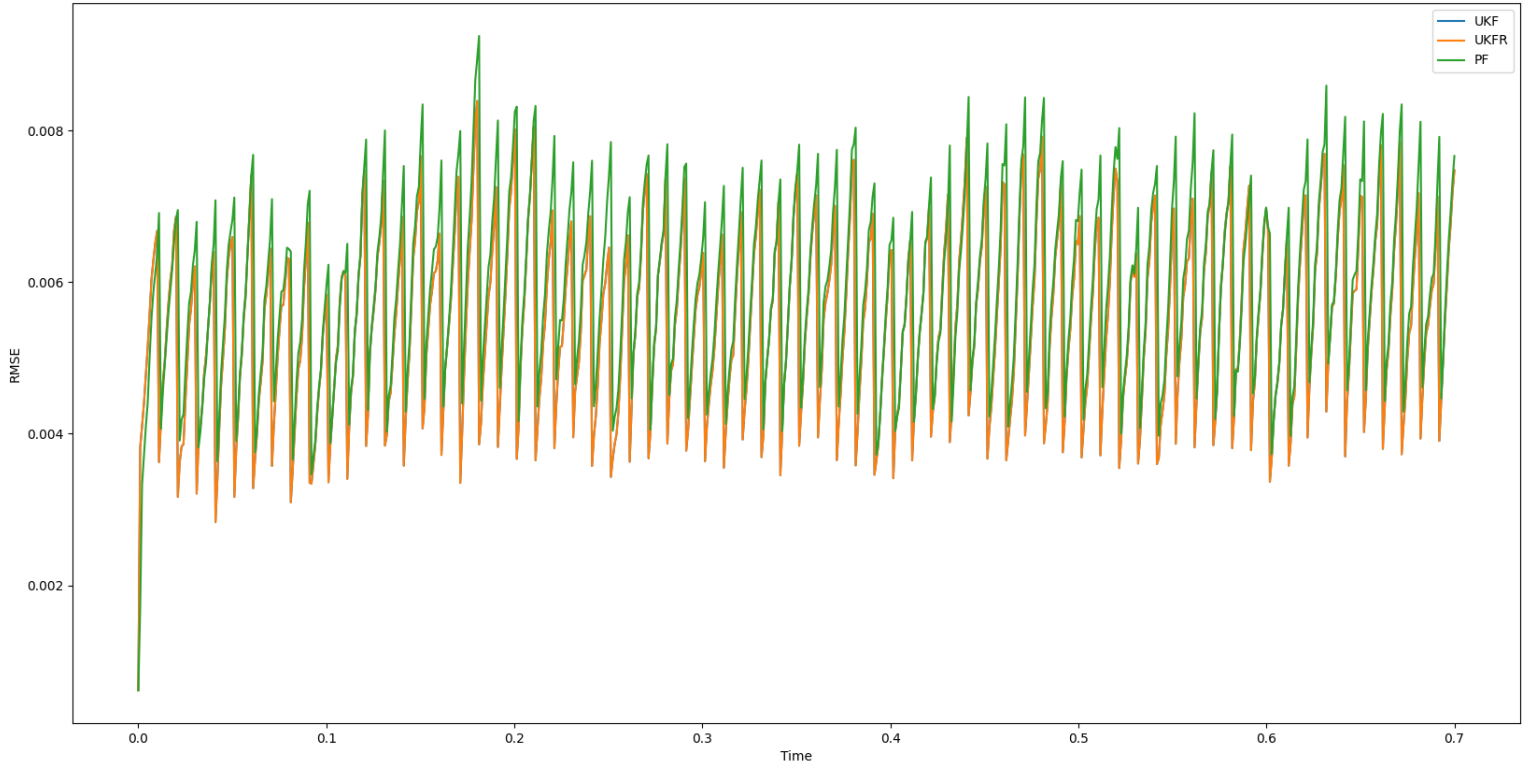


Figure 11: Error of θ

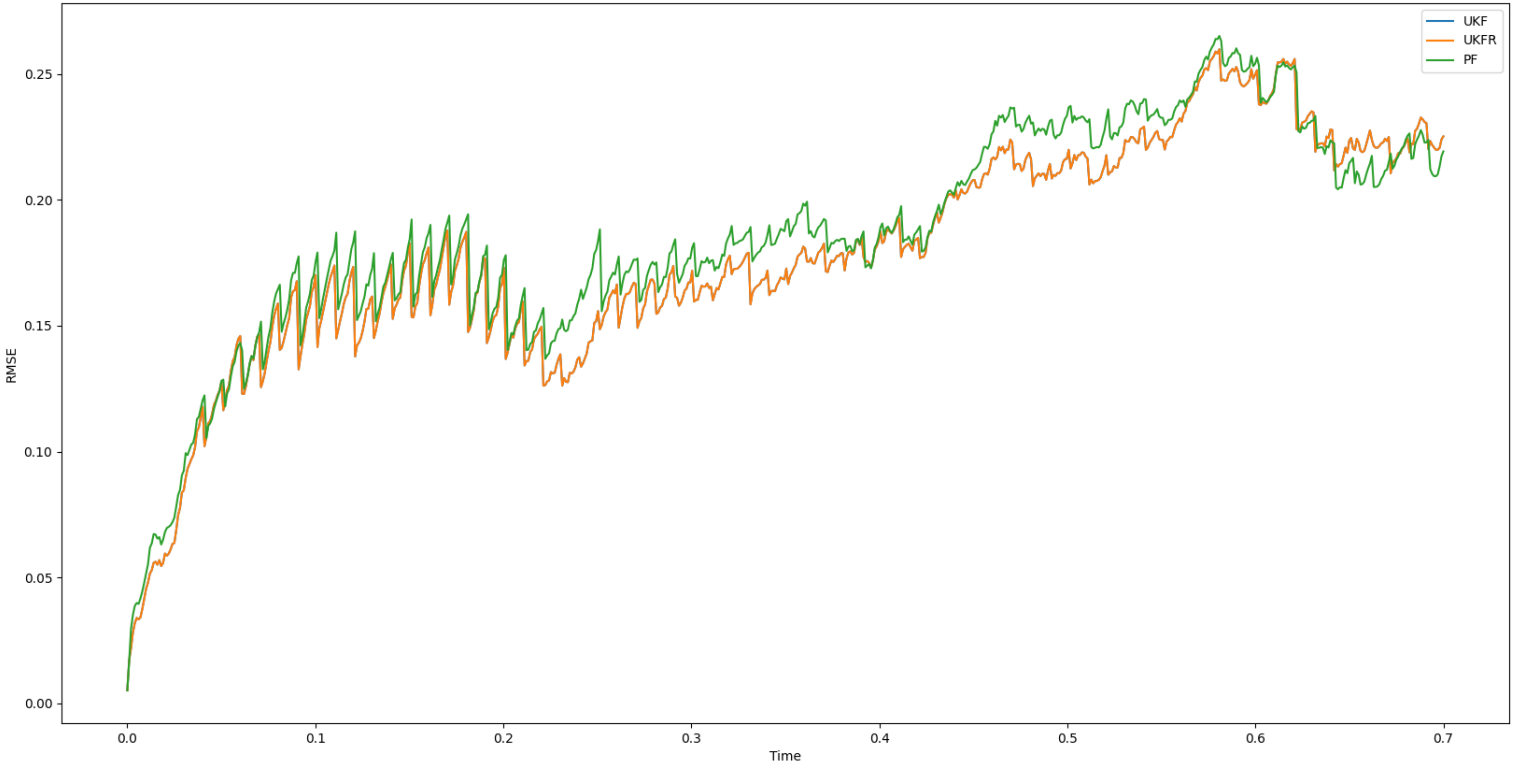
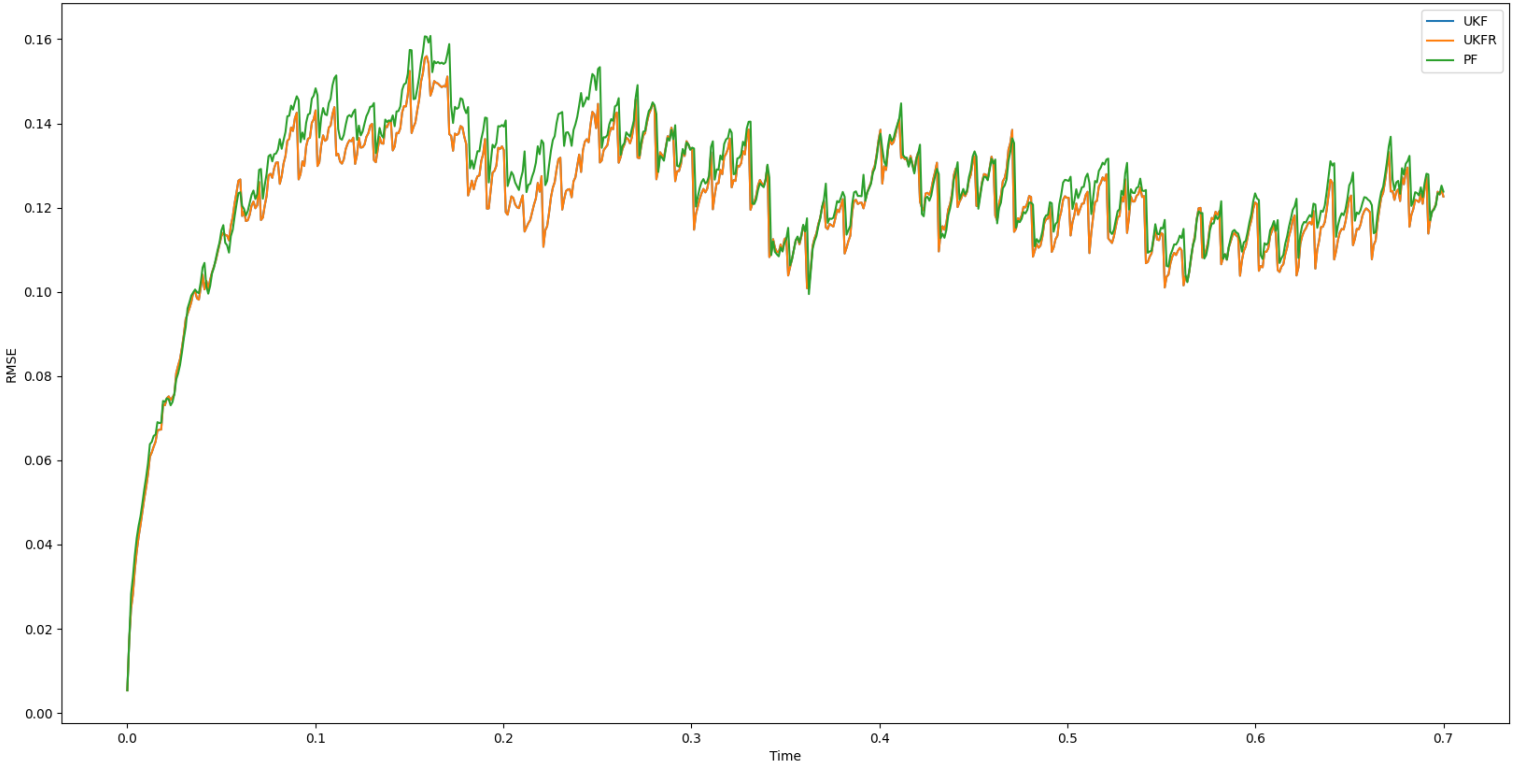


Figure 12: Error of ϕ



2.2.4 Numerical Results

Method	Percentage of Divergent Trajectories
UKF	0.0
UKFR	0.0
PF	0.0

Time	UKF	UKFR	PF
0.1	0.00581	0.00581	0.00579
0.3	0.00538	0.00538	0.00542
0.5	0.00481	0.00481	0.00488

Table 1: Average filter errors when estimating x_1

Time	UKF	UKFR	PF
0.1	0.00515	0.00515	0.00516
0.3	0.00531	0.00531	0.00549
0.5	0.00547	0.00547	0.00561

Table 2: Average filter errors when estimating x_2

Time	UKF	UKFR	PF
0.1	0.14177	0.14177	0.14786
0.3	0.14081	0.14081	0.14322
0.5	0.17747	0.17747	0.18262

Table 3: Average filter errors when estimating θ

Time	UKF	UKFR	PF
0.1	0.11516	0.11516	0.12094
0.3	0.10956	0.10956	0.10971
0.5	0.09634	0.09634	0.10019

Table 4: Average filter errors when estimating ϕ

3 Conclusions from the Calculations

The plots show that the estimates produced by the UKF and its square-root modification behave in exactly the same way. This logically follows from the implementation details described above for 128-bit numbers. More details about this behavior of Kalman filters can be found in the corresponding section of the documentation.

It can also be seen that the filters perform best when estimating the cart's position and worst when estimating the angles θ and ϕ. There is a natural explanation for this: the observations provide the cart's polar coordinates. This allows us to construct fairly accurate estimates for x_1 and x_2. At the same time, the angles do not appear in the observations in any direct way, so information about them is obtained only indirectly through the cart's motion. Moreover, since the angular velocity u_2 was taken equal to zero in the computations, the behavior of ϕ is described by a Wiener process with the corresponding parameters. Because of this, it is the hardest quantity to estimate, although it can still be inferred from the evolution of θ. However, since all observations are noisy and noise is also added when the state changes, the effect of error accumulation is most visible precisely on this state component. Nevertheless, the main task can be considered accomplished: the resulting algorithm estimates the target's motion trajectory with sufficient accuracy despite disturbances. The tables also show that the estimate obtained by the particle filter is often worse than the UKF estimate (the error is larger by approximately a factor of 1.5–1.6), but at the same time the percentage of divergent trajectories is lower for PF (6 versus almost 17). Thus, in this case the greater computational complexity is justified by greater stability. If we also look at the plots of mean errors over all trajectories and over non-divergent trajectories only, we can see that the particle filter behaves similarly on both, whereas the mean error of the sigma-point filter strongly depends on the type of trajectory.

If we look at the plots of mean errors over all trajectories, we can also notice that the mean error grows over time. This may be due to the fact that, starting from some moment, the cart may begin circling around a certain point because of the uncontrolled growth of the steering angle. At such moments, predicting the trajectory becomes difficult, since a small change in the cart's position compared with the observation error strongly affects filter performance, and from the standpoint of the problem formulation such motion becomes hardly distinguishable from Brownian motion. It is possible that this behavior of the target could be avoided by replacing constant control with time-dependent control.

4 Appendix

Consider the system of stochastic differential equations

$$\begin{aligned} \frac{dx}{dt} &= u_1 \cos\theta(t) \cos\phi(t), \\ \frac{d\theta}{dt} &= \frac{u_1}{l} \sin\phi(t) + \omega_3(t), \\ \frac{d\phi}{dt} &= \omega_4(t), \end{aligned} \tag{1}$$

with initial conditions

$$x(0) = 0, \quad \theta(0) = \frac{\pi}{4}, \quad \phi(0) = 0.$$

The white noises $\omega_3(t)$ and $\omega_4(t)$ are represented via standard Wiener processes as follows:

$$\omega_3(t)\, dt = \sigma_3\, dW_3(t), \qquad \omega_4(t)\, dt = \sigma_4\, dW_4(t),$$

where $W_3(t)$ and $W_4(t)$ are standard Brownian motions with variance t, that is,

$$\mathbb{D}\{\sigma_3\, W_3(t)\} = \sigma_3^2\, t \quad \text{and} \quad \mathbb{D}\{\sigma_4\, W_4(t)\} = \sigma_4^2\, t.$$

4.1 Analytical Solution of the System

Solution for $\phi(t)$

The equation for $\phi(t)$ is

$$\frac{d\phi}{dt} = \omega_4(t),$$

with $\phi(0) = 0$, and has the solution

$$\phi(t) = 0 + \sigma_4\, W_4(t) = \sigma_4\, W_4(t). \tag{2}$$

Solution for $\theta(t)$

The equation for $\theta(t)$ is

$$\frac{d\theta}{dt} = \frac{u_1}{l} \sin\phi(t) + \omega_3(t),$$

with $\theta(0) = \pi/4$, which yields the integral representation

$$\theta(t) = \frac{\pi}{4} + \frac{u_1}{l} \int_0^t \sin\Big(\sigma_4\, W_4(s)\Big) ds + \sigma_3\, W_3(t). \tag{3}$$

Solution for $x(t)$

The analytical solution for $x(t)$ can be written as

$$x(t) = 0 + u_1 \int_0^t \cos\theta(s) \cos\phi(s)\, ds, \tag{4}$$

that is,

$$x(t) = u_1 \int_0^t \cos\theta(s) \cos\phi(s)\, ds.$$

Substituting expressions (2) and (3) for $\phi(s)$ and $\theta(s)$, we obtain

$$x(t) = u_1 \int_0^t \cos\left\{\frac{\pi}{4} + \frac{u_1}{l}\int_0^s \sin\Big(\sigma_4\, W_4(\tau)\Big)d\tau + \sigma_3\, W_3(s)\right\} \cos\Big(\sigma_4\, W_4(s)\Big)\, ds.$$

This is the full analytical representation of the solution of the system.

4.2 Computation of the Mathematical Expectation

Denote the mathematical expectation of the process $x(t)$ by

$$X(t) = \mathbb{E}[x(t)].$$

From (4) it follows that

$$X(t) = u_1 \int_0^t \mathbb{E}\Big[\cos\theta(s)\ \cos\phi(s)\Big]ds.$$

Write $\theta(s)$ in the form

$$\theta(s) = A(s) + \sigma_3\, W_3(s),$$

where

$$A(s) = \frac{\pi}{4} + \frac{u_1}{l}\int_0^s \sin\Big(\sigma_4\, W_4(\tau)\Big)d\tau.$$

Using the independence of the processes W_3 and W_4 and the known result for a normal random variable, we obtain

$$\mathbb{E}\Big[\cos\Big(A(s) + \sigma_3\, W_3(s)\Big)\Big] = \cos A(s)\ \exp\Big(-\frac{\sigma_3^2\, s}{2}\Big).$$

Thus, the mathematical expectation can be written as

$$X(t) = u_1 \int_0^t \exp\Big(-\frac{\sigma_3^2\, s}{2}\Big)\, \mathbb{E}_{W_4}\left\{\cos\Big[A(s)\Big], \cos\Big(\sigma_4\, W_4(s)\Big)\right\}ds.$$

The final form of the mathematical expectation is therefore

$$\boxed{X(t) = u_1 \int_0^t \exp\Big(-\frac{\sigma_3^2\, s}{2}\Big)\, \mathbb{E}_{W_4}\left\{\cos\left[\frac{\pi}{4} + \frac{u_1}{l}\int_0^s \sin\Big(\sigma_4\, W_4(\tau)\Big)d\tau\right], \cos\Big(\sigma_4\, W_4(s)\Big)\right\} ds.} \tag{5}$$

4.3 Proof of the Convergence of $X(t)$ as $t \to +\infty$

Our task is to show that the limit

$$X(\infty) = \lim_{t\to+\infty} X(t)$$

exists and is finite.

Step 1. Estimate of the integrand.

Since

$$\left|\cos\left[\frac{\pi}{4}+\frac{u_1}{l}\int_0^s \sin\Big(\sigma_4\, W_4(\tau)\Big)d\tau\right]\,\cos\Big(\sigma_4\, W_4(s)\Big)\right| \le 1,$$

we obtain

$$\left|\exp\Big(-\frac{\sigma_3^2\, s}{2}\Big)\,\mathbb{E}_{W_4}\{\cdots\}\right| \le \exp\Big(-\frac{\sigma_3^2\, s}{2}\Big).$$

Step 2. Integrability of the dominating function.
Consider the function

$$g(s)=\exp\Big(-\frac{\sigma_3^2\, s}{2}\Big).$$

The integral

$$\int_0^{+\infty} g(s)\,ds=\int_0^{+\infty}\exp\Big(-\frac{\sigma_3^2\, s}{2}\Big)ds=\frac{2}{\sigma_3^2}$$

is finite.

Step 3. Application of the Lebesgue Dominated Convergence Theorem.
Since the integrand

$$f(s)=\exp\Big(-\frac{\sigma_3^2\, s}{2}\Big)\,\mathbb{E}_{W_4}\left\{\cos\left[\frac{\pi}{4}+\frac{u_1}{l}\int_0^s \sin\Big(\sigma_4\, W_4(\tau)\Big)d\tau\right],\cos\Big(\sigma_4\, W_4(s)\Big)\right\}$$

is bounded by the function $g(s)$, which is integrable on $[0,+\infty)$, the Lebesgue Dominated Convergence Theorem implies that the integral

$$\int_0^{+\infty} f(s)\,ds$$

converges to a finite number. Therefore,

$$\boxed{X(\infty)=u_1\int_0^{+\infty}\exp\Big(-\frac{\sigma_3^2\, s}{2}\Big)\,\mathbb{E}_{W_4}\left\{\cos\left[\frac{\pi}{4}+\frac{u_1}{l}\int_0^s \sin\Big(\sigma_4\, W_4(\tau)\Big)d\tau\right],\cos\Big(\sigma_4\, W_4(s)\Big)\right\}ds<\infty.}$$

4.4 Conclusions About the Trivial Trajectory Estimate

Since $X(t)$ is bounded as $x\to\infty$ ($|X(t)|<\frac{2}{\sigma_3^2}$) and converges, we may conclude that the trivial estimate for x_1 converges as $t\to\infty$. This is also confirmed by the practical results (see the estimate plots). For x_2, the proof and the conclusion are constructed analogously.